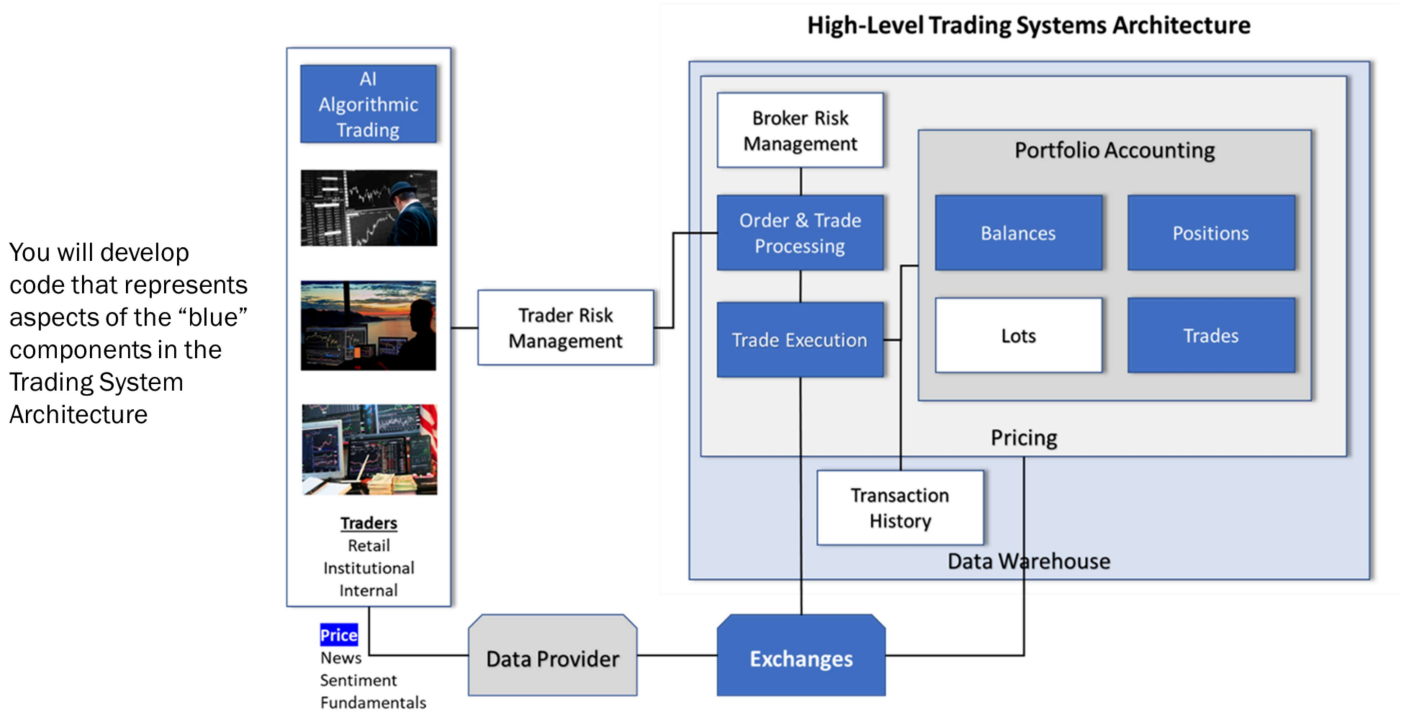
Machine Learning for Trading Project

As part of the ML4T course, we're setting you a challenging but rewarding task: constructing a fully functional, simulated trading system piece by piece. Through this hands-on project, you will bring to life the various components that form the backbone of any robust trading system. The ultimate goal is to synthesize your knowledge of machine learning and algorithmic trading strategies into a practical application that mirrors the complexity of real-world financial markets.

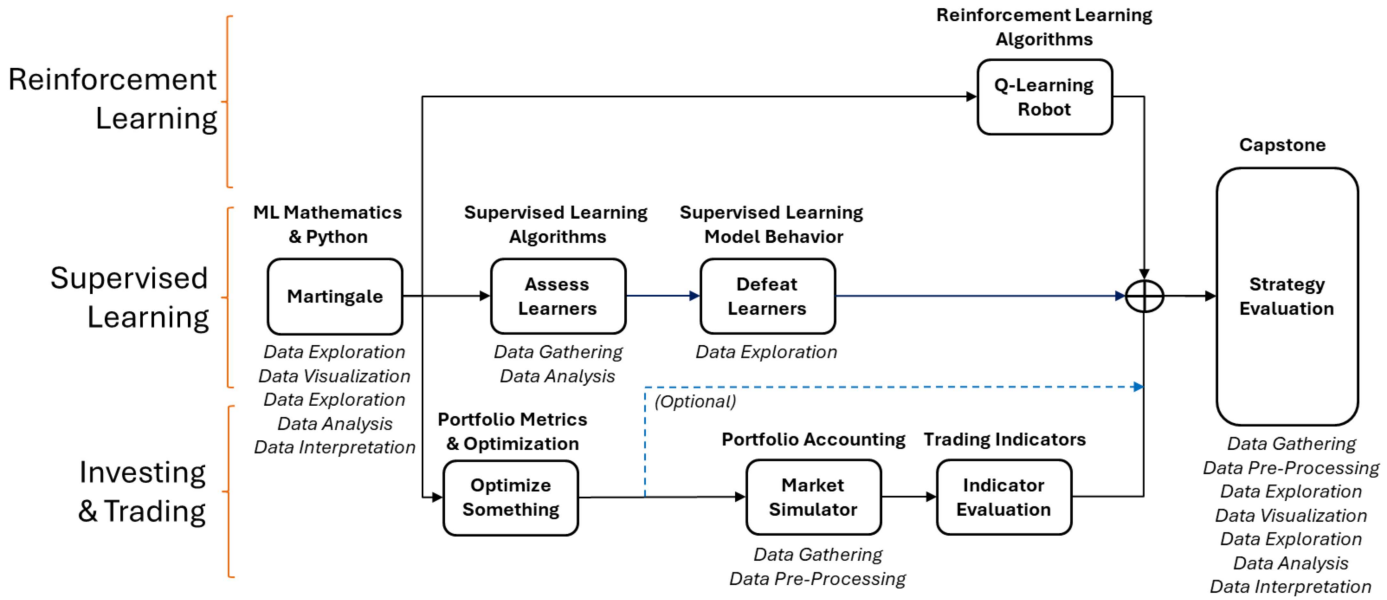
Throughout the course, you will focus on creating and integrating the elements highlighted in blue on the High-Level Trading Systems Architecture diagram.



Each piece you develop will be a learning opportunity to understand how complex systems are architected, integrated, and managed. By the end of the project, you'll not only have a deeper understanding of the theoretical underpinnings of machine learning in finance but also a tangible system that demonstrates your ability to apply those principles in practice.

Iterative Development Towards the Capstone Project

Throughout this course, we're not just jumping straight into building a complex system. Instead, we'll approach it iteratively, ensuring that with each project, you're layering on knowledge, concepts, and skills that are critical to understanding and creating a sophisticated AI trading agent.



Each project is a building block:

- **Projects P1 through P7** are carefully designed to introduce key ideas and concepts, and also to develop actual modules that will be integrated into your final creation. Think of these projects as the gears of a watch, each one essential and intricate, working together to keep precise time.
- **Project P8** is the culmination of all your hard work: the capstone project. Here, you will leverage all the components you've built and knowledge you've acquired to construct and simulate a fully operational AI Trading agent. This is where everything comes together, where theory meets practice, and where you get to see the fruits of your labor in action. By the end of Project P8, you'll have a mini-version of a trading system that employs the cutting-edge techniques you've learned throughout the course.

Understanding and Navigating Your Assignments

Each assignment is detailed on its respective Assignment page. While we strive to answer many of the questions that commonly arise, it's important to recognize that, much like in professional settings, assignment instructions may not be presented as a straightforward checklist. Success in these projects requires you to engage deeply with the instructions provided, discern the underlying requirements, and address them in your implementation. We suggest using a highlighter to mark potential requirements as you review the assignment materials.

About Pre-Validation Scripts

To aid in your development process, a pre-validation or local grading script is provided with each assignment. Please note that this script is designed for convenience and does not encompass all specific assignment requirements. For instance, if an assignment requires listing all 50 U.S. states in alphabetical order, the script might only verify the presence of 50 entries. This could mean passing the script by mistakenly repeating "Georgia" 50 times. Therefore, the pre-validation script should not be considered a comprehensive test suite but rather a preliminary check to ensure your code meets basic criteria.

Developing Your Own Tests

Given the limitations of the pre-validation scripts, you are strongly encouraged to develop your own test cases based on the requirements you identify. Creating detailed and rigorous tests for your work will not only help you meet the assignment's demands more effectively but also enhance your problem-solving skills and prepare you for real-world challenges in software development and data analysis.

Project Links

- **Assignment Instructions**:: **Assignment Overview** (https://gatech.instructure.com/courses/498126/pages/machine-learning-for-trading-project) | **Project 1** (https://gatech.instructure.com/courses/498126/assignments/2333578) | **Project 2** (https://gatech.instructure.com/courses/498126/assignments/2333582) | **Project 3** (https://gatech.instructure.com/courses/498126/assignments/2333586) | **Project 4** (https://gatech.instructure.com/courses/498126/assignments/2333588) | **Project 5** (https://gatech.instructure.com/courses/498126/assignments/2333590) | **Project 6** (https://gatech.instructure.com/courses/498126/assignments/2333592) | **Project 7** (https://gatech.instructure.com/courses/498126/assignments/2333594) | **Project 8** (https://gatech.instructure.com/courses/498126/assignments/2333598) | **API Specifications** (https://gatech.instructure.com/courses/498126/pages/api-specifications)

So, as we move through the coursework, remember that each project, each line of code, and each decision you make brings you one step closer to realizing the ultimate goal. Get ready to build, learn, and innovate!